Title: SpaceX Falcon 9 First Stage Landing Prediction

Subtitle: Applied Data Science Capstone Project

Presented by: GC Viswa prakash

GitHub Repository: https://github.com/vg9359-stack/IBM-DataScience-Capstone

Executive Summary

- **Objective:** Predict if the Falcon 9 first stage will land successfully to estimate reuse costs.
- **Data Sources:** SpaceX REST API, Wikipedia web scraping, and historical launch records.
- **Key Methodology:** Applied Data Wrangling, One-Hot Encoding, EDA via Seaborn & SQL, Interactive Visual Analytics (Folium & Dash), and Machine Learning Classification Models.
- **Key Finding:** Machine Learning models (Logistic Regression, Decision Tree, SVM, and KNN) achieved up to **83.33% accuracy** on test data.

Introduction & Problem Statement

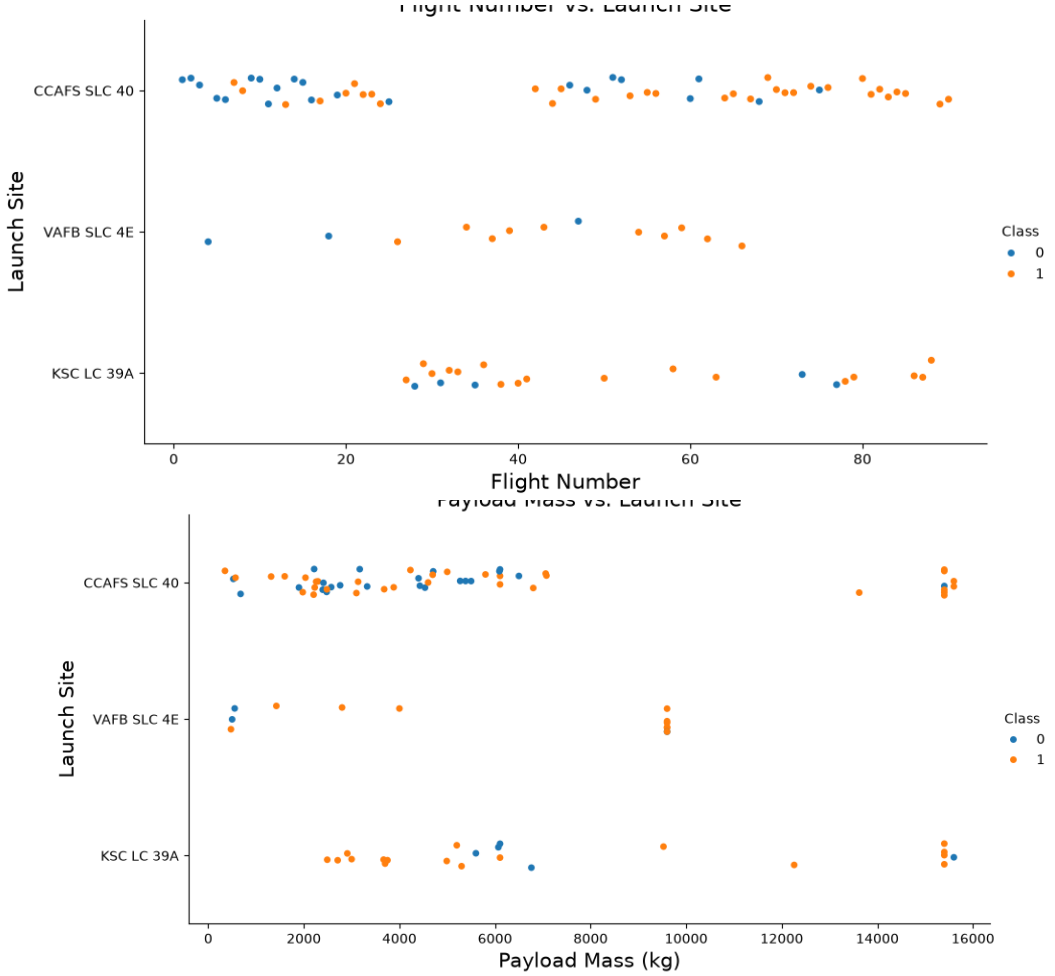
- **Background:** SpaceX advertises Falcon 9 rocket launches at $62 million; competitors cost upwards of $165 million. Much of the savings come from reusing the first stage.
- **Problem Statement:** Determine if the first stage will land successfully so competitor launch costs can be determined accurately.
- **Goal:** Train binary classification models to predict landing outcome (1 = Success, 0 = Failure).

Data Collection & Wrangling

- **API Ingestion:** Retrived launch records including payload mass, launch sites, orbit, and outcome from SpaceX API.
- **Web Scraping:** Scraped Wikipedia launch history tables using BeautifulSoup.
- **Data Preprocessing & Cleaning:**
  - Handled missing payload values using column mean imputation.
  - Converted outcomes into binary landing outcome class (1 for successful landing, 0 for failed/no landing).
- **Feature Engineering:** Applied One-Hot Encoding to categorical features (Orbit, LaunchSite, LandingPad, Serial), expanding features to **83 columns**.

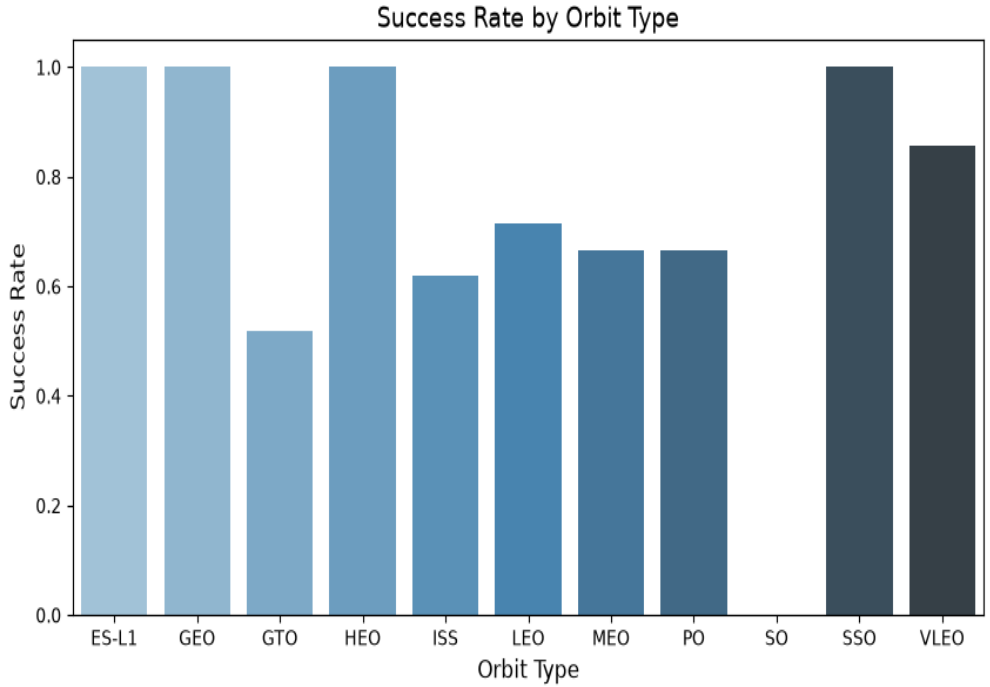
EDA - Flight Number & Payload Mass vs. Launch Site

- **Flight Number vs. Launch Site:**
  - Earlier flights were concentrated at CCAFS LC-40 with lower success rates.
  - Success rate significantly improved across all sites as flight numbers increased.
- **Payload Mass vs. Launch Site:**
  - Payloads over 10,000 kg had higher success rates.
  - VAFB SLC-4E handles lower payload counts with consistent landing outcomes.



EDA - Orbit Types & Success Rates

Highest Success Rates: ES-L1, GEO, HEO, and SSO orbits achieved near 100% success rates.
Moderate Success Rates: VLEO and LEO showed strong positive landing performance (~80%).
Lowest Success Rates: GTO orbit showed higher failure/no-landing counts due to high-altitude recovery constraints.



EDA with SQL Analysis

- **Query Insights Executed:**
- Identified unique launch sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40.
- Filtered total payload mass and minimum payload records.
- Evaluated mission outcomes grouped by launch site.
- **SQL Output Table Example:**

Launch Site	Outcome Count
CCAFS LC-40	26
VAFB SLC-4E	13
KSC LC-39A	10
CCAFS SLC-40	34

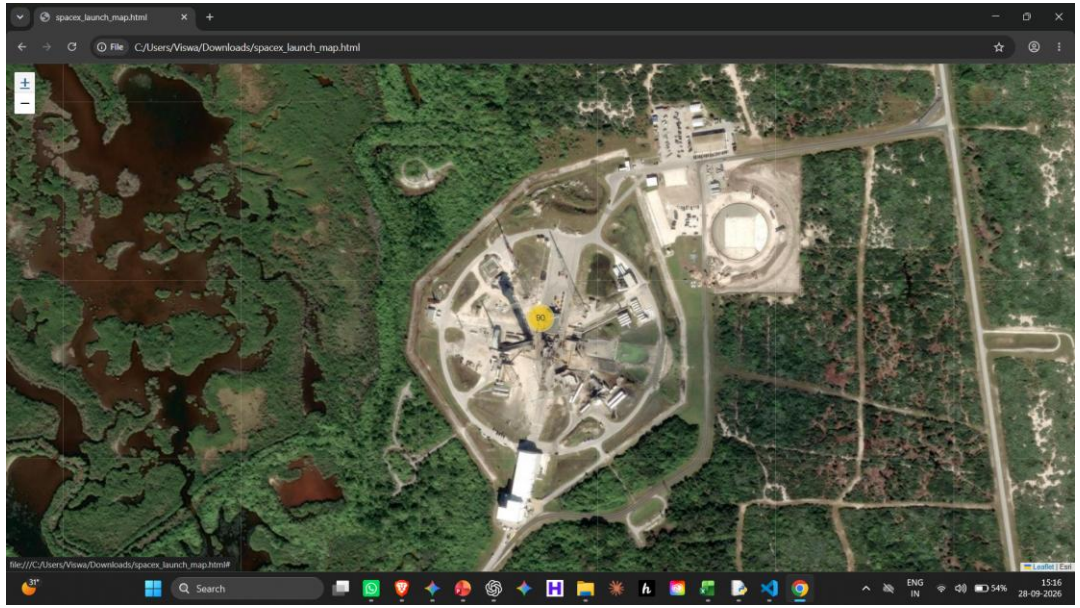
Interactive Visual Analytics (Folium & Plotly Dash)

Folium Geospatial Analysis:

Plotted launch site markers with color-coded outcomes (Green = Success, Red = Failure).
Calculated distances to coastlines, railways, and highways to analyze proximity risks.

Plotly Dash Application:

Dynamic dropdown filter for individual vs. total launch site success distribution.
Interactive payload range slider dynamically re-rendering scatter plots.



Machine Learning Models & Results

- **Dataset Splitting:** 80% Training Data, 20% Test Data (**18 Test Samples**).
- **Feature Scaling:** Applied StandardScaler to ensure uniform feature weight across distance-based models.
- **Hyperparameter Tuning:** Executed GridSearchCV with 10-fold Cross-Validation across 4 algorithms.

Classification Results Table

Model	Best Hyperparameters / Kernel	Test Accuracy
Logistic Regression	solver='lbfgs', C=0.01	83.33%
Support Vector Machine (SVM)	kernel='sigmoid'	83.33%
Decision Tree Classifier	criterion='gini', max_depth=4	83.33%
K-Nearest Neighbors (KNN)	n_neighbors=4	83.33%

Conclusion & Business Recommendations

- **Model Conclusion:** All four tuned classification models achieved an identical top score of **83.33% test accuracy**.
- **Key Drivers:** Payload mass, orbit type, and booster flight counts are the strongest predictors of landing success.
- **Business Insight:** As SpaceX continues reusing boosters (higher flight counts), landing predictability approaches near-certainty, allowing competitors to confidently bid lower on launch contracts using reusable rocket technologies.